

RhoA Protein in Genetically Compromised SERT Rats

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Olivia McGifford

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Abstract

The current study aimed to assess changes in the small GTPase, RhoA using the Serotonin Reuptake Transporter Knock Out (SERT KO) animal model. A Western Blotting procedure was developed and optimised to determine RhoA protein levels in wildtype, heterozygous and homozygous SERT KO rats. Using densitometry analysis and ANOVA, a significant difference in RhoA relative densities was found between wild types and homozygous SERT KO rats. However, no significant difference was found between heterozygous SERT KO rats and either wild types or homozygous SERT KO rats. Although the sample size was small, and this study should be regarded as a pilot, the results confirm our hypothesis, SERT KO rats have increased RhoA protein levels compared to wild types. Increased RhoA protein levels are likely to affect the development of the brain and contribute to behavioural alterations, such as increases in anxiety and depression like symptoms, and enhanced sensitivity to drugs of abuse. Further optimisation of RhoA western blotting protocol is necessary, and more samples should be tested. In addition, RhoA levels should be assessed early in life.

RHOA PROTEIN IN GENETICALLY COMPROMISED SERT RATS

1.1. Neuropsychiatric Disorders and Serotonin

The development of neuropsychiatric disorders can be attributed to many factors in and individual's life such as genetics, biological phenomena, and environmental events throughout childhood and adulthood. The main characteristics of neuropsychiatric disorders include continued occurrence of psychiatric symptoms, cognitive impairments, early cerebral symptoms and often times imbalances in brain biochemistry (Miyoshi, Morimura, Maeda, 2010). An imbalance in brain chemistry has been implicated as one of the key factors in neuropsychiatric disorder development (Terry, Buccafusco, Wilson, 2008). The neurotransmitter serotonin (5-HT) has been identified as playing a key role in brain function, and 5-HT imbalances are well known to have effects on neuropsychiatric disorder development (Terry et al., 2008). 5-HT is held within vesicles located in the presynaptic neuron. When an action potential fires 5-HT is released from the vesicles/presynaptic neuron into the synaptic cleft. 5-HT then binds to specific 5-HT receptors, leading to activation or inhibition of the postsynaptic cells. 5-HT is then either degraded via monoamine oxidase enzymes (Gingrich, Hen, 2001), or alternatively 5-HT is replenished to the vesicles via reuptake through the Serotonin Reuptake Transporter (SERT), located on the presynaptic neuron. SERT plays a crucial role in the termination of serotonergic transmission through a sodium dependent uptake of 5-HT, and can mediate extracellular levels on 5-HT present in the synaptic cleft (Lesch et al., 1994).

The human SERT protein is encoded from a single gene called SLC6A4. This gene is located on chromosome 17q 11.1 - 17 q12 and consists of fourteen exons (Margoob, Mushtag, Murtza, Mushtag, 2008). SERT has been identified as a key component in the regulation of 5-HT and is one of the main targets of neuropsychiatric disorder treatment (Gingrich, Hen, 2001). Many polymorphisms of SERT in humans have been identified,

however the most well-known is 5-HTTLPR. The 5-HTTLPR polymorphism consists of a 44-base pair deletion or insertion in the promoter region, which consequently results in either the 'long allele' or 'short allele' of the SERT gene (Jawahar, Toben, Baune, 2019). The short allele is associated with decreases in SERT expression, whereas the long allele is associated with normal levels of SERT expression (Jawahar, Toben, Baune, 2019). There are many other variants of SERT polymorphisms, however these are less common and moreover, are less investigated and understood. Polymorphisms of SERT have been both negatively and positively associated with many neuropsychiatric disorders such as depression, anxiety, OCD and PTSD (Margoob et al., 2008). Changes in level of SERT expression can have cascading effects, resulting in the increase or decrease in serotonergic transmission and levels of 5-HT. Fluctuations in the levels of 5-HT may have mediating effects on further processes within the brain, possibly leading to development of neuropsychiatric disorders (Jawahar, Toben, Baune, 2019).

Significant changes and imbalances in 5-HT levels, receptor densities and neuronal activity have been observed in individuals with neuropsychiatric disorders such as anxiety and depression. Through studies in patients with anxiety and depression, and use of psychoactive drugs targeting SERT, researchers have been able to identify the importance of 5-HT and SERT in the development of these neuropsychiatric disorders (Gingrich, Hen, 2001). Selective Serotonin Reuptake Inhibitors (SSRIs) are prescribed to patients who are diagnosed with anxiety and/or depressive disorders as a form of medicinal treatment (Terry et al, 2008). SSRIs aim to reinstate the normal balance of 5-HT in the brain through the blockade of the SERT protein. Blockading of SERT leads to an increase in 5-HT in the synaptic cleft and therefore extended frequency of 5-HT receptor activation (Gingrich, Hen, 2001). SSRIs alleviate symptoms of anxiety and depression, and through these consequences

researchers are given insight into the importance of SERT and 5-HT in neuropsychiatric disorders.

Individuals with a genetic polymorphism of SERT are at higher risk of developing neuropsychiatric disorders (Gingrich, Hen, 2001). However, this seems contradictory to the function of SSRI treatment. In both circumstances, there is a decrease in SERT compared to normal, conversely SSRIs alleviate symptoms, and SERT polymorphisms increase the likelihood of symptom development. This particular phenomenon is known as the serotonin paradox. Decreased SERT expression and SERT blockage function similarly, however seem to generate opposite consequences. A potential explanation of this paradox may lie in the fact that a genetic reduction of SERT leads to increases of extracellular 5-HT during development (Gingrich, Hen, 2001). 5-HT and its related receptors play a crucial role in the development of the brain and synaptic/neuronal plasticity. Indicating a SERT polymorphism mediated chronic increase in 5-HT, may have developmental implications on the brain leading to increased likelihood of disorder development. This role in development and plasticity could be a potential explanation for the serotonin paradox mentioned above. Thus, we would expect a genetic reduction of SERT leads to plasticity and developmental changes in the brain from very early on. However, this is very difficult to study in humans, instead animal models have been developed to help such research

1.2 Animal Models

Animal models have been used in psychology to assess the effects of genetics, neurobiology and environmental changes. Animal models are used as a function to research and relate back to human behaviour and biology. Animal models aim to bring forth insight into how certain genes and environments can affect the development of the brain (Miyoshi et al., 2010). A common animal model used to study neuropsychiatric disorders such as anxiety and depression is the SERT knock out (SERT KO) model, in which rats are bred to have

genetic changes in expression of SERT (Olivier et al., 2008). We cannot definitively say these animals are diagnosed with depression or anxiety, however they demonstrate symptoms similar to those seen in depressed and anxious patients (Olivier et al., 2008) Rats in the SERT KO model are bred to either present with heterozygous (+/-) or homozygous (-/-) SERT gene knock out. Heterozygous animals present with approximately 50% SERT expression, while homozygous have 0% SERT expression (Olivier et al, 2008). The model then uses wild type (+/+) rats who have normal expression of SERT to compare to heterozygous and homozygous SERT KO rats. Studies have demonstrated heterozygous SERT KO rats have elevated levels of 5-HT compared to wild types, and, homozygous SERT KO rats have further elevated levels of 5-HT (Homberg et al., 2007). Elevated levels of 5-HT in both heterozygous and homozygous SERT KO rats is due to the decreased expression of the SERT gene, allowing for 5-HT to be present in the synaptic cleft for longer periods of time before being metabolised. Therefore, rats with the homozygous SERT KO gene have the highest levels of 5-HT due to having no SERT expression resulting in complete inability for 5-HT to be metabolised via SERT.

As discussed earlier, changes in levels of 5-HT and 5-HT receptor activation may lead to developmental consequences in the brain, neuropsychiatric disorders and changes in synaptic plasticity.

1.3 Rho GTPases and RhoA

From the above literature, we can see 5-HT plays a key role in the development and changes in the brain and demonstrates the importance of 5-HT and how it may have an effect on synaptic plasticity and the ability for the brain to adapt to changes. However, it is thought elevated levels of 5-HT may have regulating effects on other key players in cortical function (Millan, Marin, Bockaert, la Cour, 2008). One of those is the effect on the Rho GTPase proteins. The Rho GTPases are structurally and biochemically simple proteins which switch

between two states; the GTP-bound form, which is regarded as the active form, and the GDP-bound form which is regarded as the inactive form (Manser, 2005). Rho GTPases are a family of proteins with regulatory properties in a number of neurodevelopmental processes such as neuronal migration, synaptogenesis, axon growth and maintenance, and dendritic arborisation and control these processes through functioning at specific locations within neurons at differing developmental stages (Tolias, Duman, Um, 2011).

However, the RhoA protein is of particular interest. RhoA is a small GTPase protein in the Rho family, and is responsible for dendritic spine retraction and loss, cytoskeletal rearrangement and cytokinesis (Tolias et al., 2011). Dendritic spine growth and retraction is evoked by extracellular molecules, such as RhoA, and are key biological mechanisms fundamental to the nervous system as well as development of synaptic plasticity (Manser, 2005). In the presence of increased RhoA, increases in dendritic spine retraction and loss occurs (Manser, 2005). Cytoskeletal rearrangement is also regulated through many different cellular processes including the activation of RhoA (Manser, 2005). Cytoskeletons consist of three filaments actin filaments, microtubules and intermediate filaments. RhoA has mediating effects on cytoskeletal rearrangement through formation of stress fibres and focal adhesions, as well as neurite retraction and cytokinesis (Manser, 2005). If cytoskeletal rearrangement is hindered through over activation of RhoA, this may lead to structural changes in neural networks causing functional disconnections and causing detriment to developmental processes.

5-HT has regulating effects on the level and activation of RhoA. This process occurs through the 5-HT₄ and 5-HT₇ receptors, which are receptors found in the highest density in the limbic system including the frontal cortex, amygdala, and hippocampus (Terry et al., 2008). The 5-HT₄ and 5-HT₇ receptor subtypes have been identified as having an important functional response (Terry et al., 2008). The activation of 5-HT₄ and 5-HT₇ receptors have

regulating effects on the RhoA protein. This occurs through G protein activation via G13 for 5-HT₄ and G12 for 5-HT₇ (Millan et al., 2008). The g-coupled protein activation of RhoA into its GDP bound form allows RhoA to engage in its respective functions. Therefore, in events in which 5-HT₄ and 5-HT₇ receptor activation is increased, RhoA activation is also further increased (Millan et al., 2008). Moreover, due to 5-HT being a regulator of RhoA (Millan et al, 2008) it can be predicted, increases in levels of 5-HT due to the genetic reduction of SERT may have upregulating effects on RhoA, causing activation of dendritic spine retraction, neurite retraction and effects on cytokinesis which may cause further dysfunctions in synaptic plasticity.

1.4 RNA sequencing and RhoA

Unpublished data from the Ellenbroek lab at Victoria University of Wellington examined levels of mRNA expressed RhoA in a SERT KO model using MDMA treated and untreated (clean) rats. Results indicated a significant increase in mRNA RhoA expression in homozygous SERT KO rats compared to wildtype in the clean condition. However, no significant differences were observed between heterozygous SERT KO rats and either wildtypes or homozygous SERT KO rats (see figure 1). Furthermore, in both wildtype and homozygous SERT KO rats in the MDMA conditions there was a downregulation in mRNA expressed RhoA compared to clean conditions (see figure 1). MDMA affects 5-HT transmission in the brain (Parrot, 2002). Serotonergic changes have been observed in animal research with MDMA, showing dose-dependent decreases in levels of 5-HT after prolonged MDMA exposure (Parrot, 2002).

Due to the decrease in 5-HT in MDMA treated rats, this would explain the downregulation in mRNA RhoA expression in wild type and homozygous rats in the RNA sequencing data. 5-HT mediates the level of RhoA, due to MDMA mediated downregulation

of 5-HT receptors, we would expect to see decreased activation, and therefore decreases in RhoA, which is illustrated in figure 1.

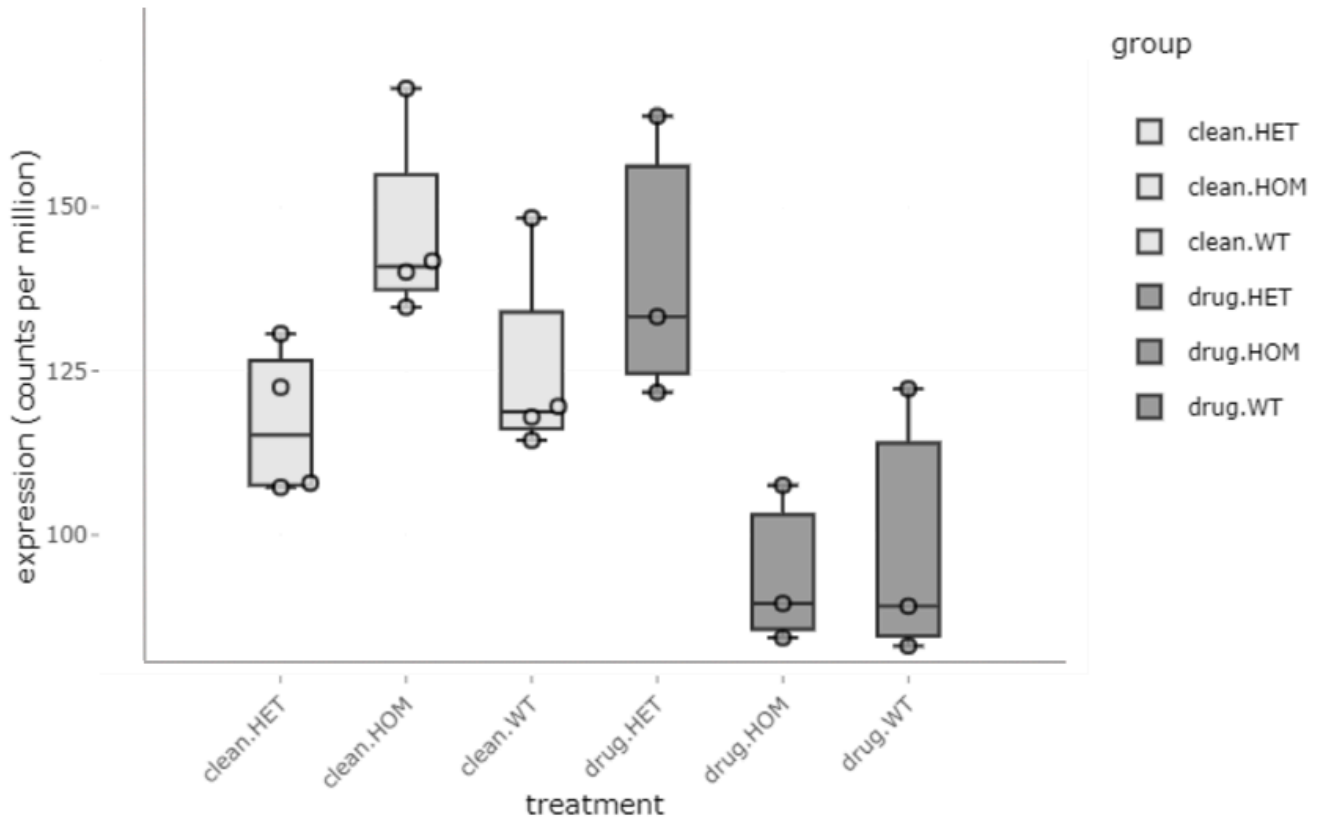


Figure 1. Box and whisker plots of RNA sequencing data of mRNA RhoA expression. Orange boxes indicated untreated, left to right; heterozygous SERT KO, homozygous SERT KO, wild type. Blue boxes indicate MDMA treatment, left to right; heterozygous SERT KO, homozygous SERT KO, wild type.

1.5 Summary

Based on the evidence above we can assume SERT polymorphisms will have regulating effects on the level of 5-HT present in the synaptic cleft. If 5-HT is increased due to a SERT KO gene, this will lead to increased extracellular 5-HT and further activation of both 5-HT₄ and 5-HT₇ receptors (Millan et al., 2008). Furthermore, increased receptor activation will increase the levels of RhoA and its respective actions and may ultimately affect cortical development including dendritic spine retraction and loss, cytoskeletal rearrangement and cytokinesis (Manser, 2005). Decreases in dendritic spines and disrupted cytoskeletal rearrangement may have consequences of cortical development including

synaptic plasticity and increase the likelihood for neuropsychiatric disorder manifestation (Chen et. al., 2012). Subsequently, it is logical to measure levels of RhoA protein in wildtype, heterozygous and homozygous SERT KO rats to assess the possible implications of 5-HT and its regulating effects on RhoA. Moreover, increased levels of the RhoA protein may lead to detrimental effects on dendritic spines, neural networks, cortical development and synaptic plasticity. Through RhoA's mediating effects of these processes, RhoA may also be implicated in the development of neuropsychiatric disorders.

1.6 Aims and Hypothesis

The main focus of this study is to develop a method in which RhoA protein levels can be measured in the frontal cortex. A Western Blotting protocol will be used and optimised in order to make these assessments. Once the Western Blotting protocol is optimised, the study aims to assess the levels of RhoA protein in the frontal cortex across three differing genotypes; wild type, heterozygous and homozygous SERT KO rats. The frontal cortex is the region of interest due to its high density of 5-HT₄ and 5-HT₇ receptors (Terry et al., 2008).

Due to increased extracellular 5-HT affecting increases in levels of RhoA we hypothesise homozygous SERT KO rats will show the highest increases in levels of RhoA compared to the heterozygous SERT KO and Wild Type rats.

Method

2.1 Animals

Nine adult Wistar rats ranging from PND 115 to 174 were used in this study. Experimental groups consisted of three different genotypes; wild types, SERT KO +/-, and SERT KO -/-. Animals were bred in the vivarium at Victoria University of Wellington and were housed in groups of three in individually ventilated polycarbonate cages. The colony temperature was maintained at 19-22°C and humidity was approximately 55%. Food and water were available ad libitum. Animals were kept in home cages until needed for

extraction. Animals used in this experiment were males. Ethical approval was granted from the Victoria University of Wellington Animal Ethics Committee; AEC25766.

2.2 Tissue Extraction and Preparation

Rats were euthanised using CO₂ asphyxiation. Immediately following euthanasia animals were decapitated. Brains were extracted and placed into the brain block (Figure 2). A razor blade was placed at the olfactory bulb and posterior points of the brain to secure. Following, one razor blade was placed at the fourth partition, then the fifth partition and sliced (see figure 3A). Brain sections were placed onto a petri dish (see figure 3B), frontal cortex was separated from subcortical regions and tissue was put into 1.5ml Eppendorf tubes containing 300µl solution of radioimmunoprecipitation assay (RIPA) buffer. To prevent degradation of the proteins, RIPA solution contained Halt™ Protease Inhibitor (PI). RIPA + PI solutions were continuously kept on ice. Following, samples were transferred into dry ice to snap freeze. Samples were kept on ice until transferred and stored at -80°C until needed at a later time.

Once required, tissue samples were thawed on ice. Samples were further homogenised by using pestles to break up large chunks of tissue. Samples were then sonicated on ice using a SONOPLUS Mini20 Ultrasonic Homogeniser to break up any remaining chunks of tissue or debris. Samples were sonicated at 15 second intervals at 3 Watts for 5 min, including a 30 second rest interval between every 15 seconds to prevent the samples from overheating. Following sonications, samples were centrifuged at 16000-16500xg for 20 minutes at 4°C to separate out any cellular debris. Once centrifuged the supernatant was removed through careful pipetting and placed into new tubes. Cellular debris was discarded. Samples were kept on ice at all times. Protein samples were then moved and stored at -20°C and could be thawed and used at any time.

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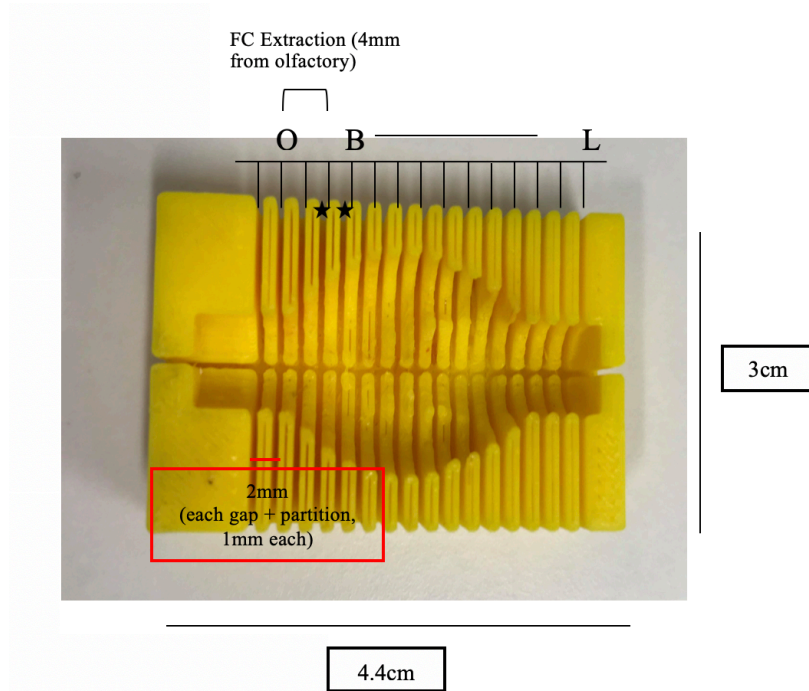


Figure 2. Brain block used as method for frontal cortex dissection and separation from the rest of the brain. 'O' indicates position of olfactory bulb in brain. B indicates Bregma line of the brain. The two star shapes indicate the positions in which razor blades were placed.

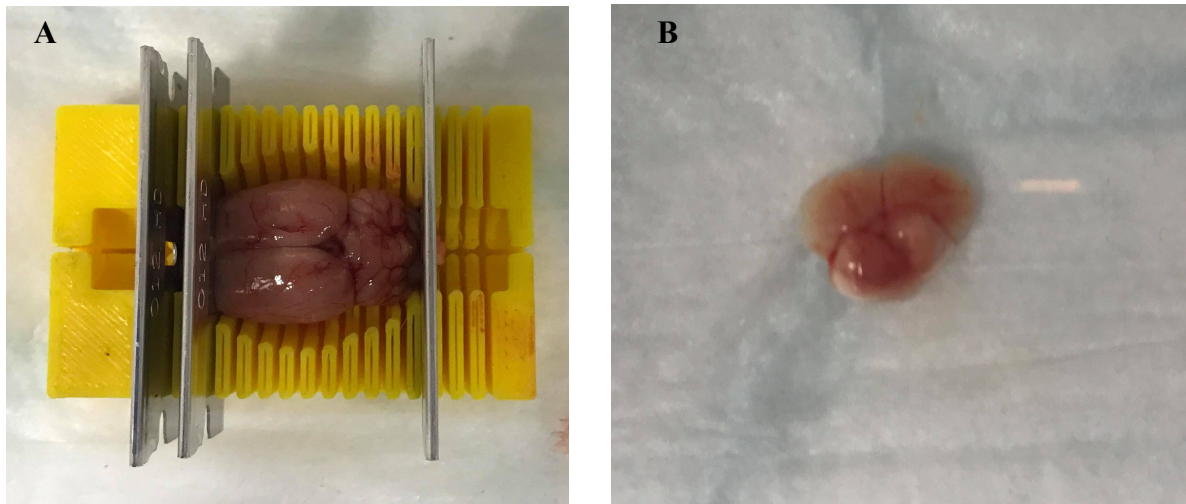


Figure 3. Dissection of the frontal cortex post extraction. A) Extraction brain placed into brain block. Posterior and anterior razor blades placed. Second razor blade from left indicated in one of the 'star' positions. B) The 2mm cortical slice of brain. Frontal cortex is two upper regions of the brain slice. Frontal cortex collected by removing tissue at a 45° angle from the midline of the slice.

2.3 Bicinchoninic Acid (BCA) Assay

A BCA assay was conducted to determine overall protein concentration in each sample. Concentrations were then used later in calculation for the Western Blotting procedure. Nine BCA standards labelled 'A - I', nine protein samples, and one 'blank' which only included the diluent were all prepared. The diluent used for both the standards and

samples was 0.1% SDS. Whilst performing the BCA, samples were diluted at a 1:9 ratio (10µl of sample: 90µl of diluent). Each standard or sample solution was pipetted in duplicate into a well within a 96 well plate. The BCA working reagent was mixed with 10ml of reagent A and 200µl of reagent B. The multiple pipette tip tool was then used to quickly add 200µl of working reagent into each well. The 96 well plate was covered and placed in an incubator at 37°C for 30 minutes. After 30 minutes of incubation the working reagent will have sufficiently reacted with standards and samples. The colour of standards and samples ranged from a light green to dark purple.

2.4 Western Blot

The Western Blotting procedure took three days. Day one included preparation of both 10% separating and 4% stacking gels. However, the 10% separating gel could be made in advance, and in this case, was kept in glass plates, wrapped in a paper towel, submerged in 1x SDS running buffer and kept at 4°C until required. A solution of protein sample, 0.1% SDS and 2x Laemmli Buffer (which contained beta-mercaptoethanol) was created for sample preparation. Using the results from the BCA assay the protein sample amount was calculated equalling to 20µg of protein per lane. A 30µl stock sample solution was made up for each sample. A 42µl ladder solution was also prepared comprising of Precision Plus Protein Dual-Colour Standard, 2x Laemmli Buffer and 0.1% SDS. All solutions were then denatured by heating at 95° for 5 minutes and following were centrifuged to ensure the solution was mixed and at the bottom of the Eppendorf tube.

Preparation for electrophoresis followed. Gels were placed facing inward into the electrophoresis tank filled with SDS-PAGE running buffer. 1.5mm, 15 well combs were then carefully removed. 10µl of ladder solution was pipetted into intended well. Following, 20µl of protein sample of interest was pipetted into its anticipated well. Once completed electrophoresis occurred for 90 – 120 minutes at 120V. During this time equipment was

prepared for the gel to membrane transfer and transfer buffer (see appendix C) was mixed and chilled for 20 minutes. Once electrophoresis was complete, the gels were quickly moved into the transfer sandwiches in which they were placed against a PVDF membrane for the proteins to transfer onto. Once the transfer tank was prepared and sandwiches were placed in the tank with the grey side facing to the back, the tank was taken into a 4°C cold room in which the transfer ran for 17 hours at 20V.

Day two consisted of the completion of the transfer. Once complete the membranes were removed and the left-hand corner was snipped to indicate the membrane was facing down. The membranes were then washed in 1x Tris buffered saline (TBS) for five minutes. Membranes were then blocked in 5% Milk Powder, 1x Tris buffered saline with Tween (TBST) solution for 60 minutes, rocking at 15-25RPM. Subsequently, Using the primary antibody, RhoA1B12 at a 1:2000 concentration in 1% milk powder/1xTBST solution, membranes were incubated overnight at 4°C rocking at 15-25RPM. A negative control membrane was also used, this was incubated in a 1% milk powder/ 1xTBST solution, consisting of no primary antibody. Coomassie staining also occurred using an SDS PAGE gel fixative and Coomassie brilliant blue powder to observe if proteins had successfully transferred.

Day three consisted of incubating membranes in the secondary antibody and scanning. Membranes were incubated for two hours at room temperature in Alexa 488 goat-anti-mouse at a 1:4000 dilution in a 1% milk powder/1xTBST solution. Negative control membranes were also incubated in secondary antibody solution. Following incubation, scanning occurred. Due to Alexa 488 being used the membrane was placed face down and scanned at 473nm FITC channel. After scanning membranes were then re-incubated in alpha-tubulin primary antibody, a 'loading control' which is used to determine equal protein loading (~20µg) across samples. A 1:10,000 concentration of alpha-tubulin was used and

primary membranes (not negative control) were incubated for 1 hour in 1% milk powder/1xTBST solution. All membranes were then incubated in a secondary antibody of Alexa 647 donkey-anti-mouse at a 1:4000 concentration in 1% milk powder/1xTBST solution for two hours rocking at 15-25rpm at room temperature. Following secondary incubation, membranes were placed face down and scanned at 635nm Cy5 laser. Alpha tubulin loading should be the same across all samples. Once scanned and satisfactory membranes were then dried out and stored in case of later use.

2.5 Protocol Optimisation

A significant amount of time in this experiment was dedicated to optimisation of the protocol. Multiple attempts of Western Blots were conducted, with minor changes being made in order to reach this optimisation. Firstly, Western Blots were run with test samples, before using experimental samples. The first Western blots run, scanned blank and with wavy ladders (see figure 7). This was assumed to be because RIPA + PI was originally being used in the sample solutions, and was causing this dysregulation. These scans also showed no sign of protein for either RhoA or alpha tubulin (loading control). It was then realised calculations for protein loading had been too low. Further western blots were conducted to continue optimisation. Test run two (see figure 8) did not show ladders or RhoA protein at 435nm FITC channel scan. However, ladders were present at the 635nm Cy5 channel scan, but no alpha tubulin was observed. It was later understood protein calculations were still incorrect and less than 10µg of protein was being loaded. Test run three consisted of using loading concentrations between 10µg and 40µg to assess if loading concentrations were correct. Both RhoA and alpha tubulin protein lanes were visible, however were smudged down the membrane (see figure 10). This indicated protein loading had been corrected. However, on negative control membranes protein *was* visible. This is thought to be researcher error by adding primary antibodies to both membranes. A further western blot protocol was conducted

with adjustments. However, electrophoresis of the protein samples did not run correctly and the proteins were smeared across the membrane, and were not situated in place (see figure 9). It was later realised beta-mercaptoethanol had not been used during sample preparation, and therefore denaturing of proteins, and protein migration did not occur properly, leading to the smearing. The final western blot procedure with all adjustments taken into consideration, lead to all nine lanes of protein showing up for both RhoA and alpha tubulin, without wavy bands or smearing, and this was used as measurable results (see figure 11&12)

2.6 Densitometry Analysis

Firstly, densitometry analyses were conducted in order to obtain values for RhoA and alpha-tubulin area and percentages. Using ImageJ, a software used for densitometry analyse for Western Blots, the protein band measurements could be obtained. The membrane scans are uploaded into ImageJ and edited to an 8-bit image (black and white) for better analysis. Using the rectangle tool, a rectangle is dragged across the first lane, then using “analyse > gels > select first lane”, the first lane is created (see figure 4). The same rectangular box is then moved and placed over subsequent lanes using “analyse > gels > select next lane” until all nine lanes are labelled (see figure 5). The lanes are then plotted onto separate graphs all showing peaks using “analyse > gels > plot lanes”. Peaks are then closed off using the straight-line tool, dragging the line from the bottom of each side of the peak (see figure 6). Once all peaks are closed off, using the wand tool, select each peak, and the area of each peak is given. Then using “analyse > gels > label peaks” the percentage of each peak is given. The area and percent of each sample is then exported into excel for further analyses (see appendix A). This process was repeated for alpha-tubulin scans in order to obtain area and percentage of alpha-tubulin loading in each of the samples.

2.7 Statistical Analysis

To calculate the density of RhoA, the percent of a sample is divided by its corresponding alpha tubulin percent to obtain the relative densities for all nine samples (see table 1). A one-way analysis of variance (ANOVA) was conducted using Jamovi statistics software. The relative densities were used as the dependent variable, and the genotypes were used as the grouping variable in order to obtain F and p values. A Levene's test was conducted to test for equality of variances. If the ANOVA is found to be significant a post-hoc Tukey test will be conducted to obtain differences between groups. To ensure there were no significant differences in loading of protein across samples, an ANOVA was conducted measuring differences alpha-tubulin percentages.

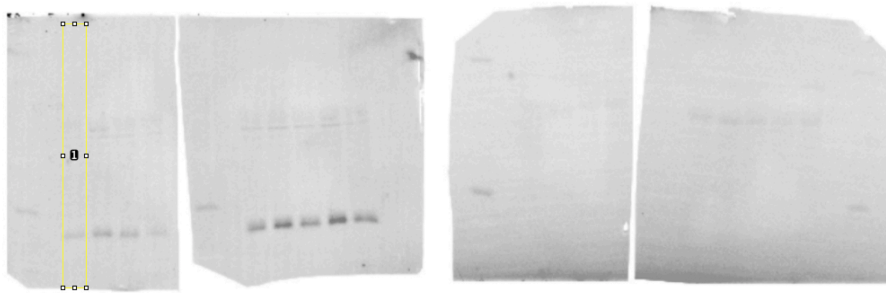


Figure 4. Rectangle tool selecting the first lane for densitometry analysis.

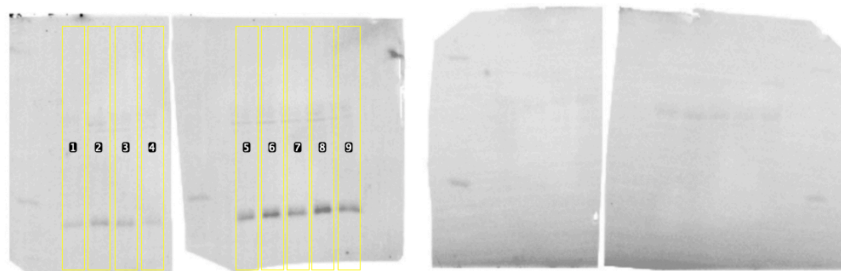


Figure 5. Rectangle tool selecting all nine lanes for densitometry analysis.

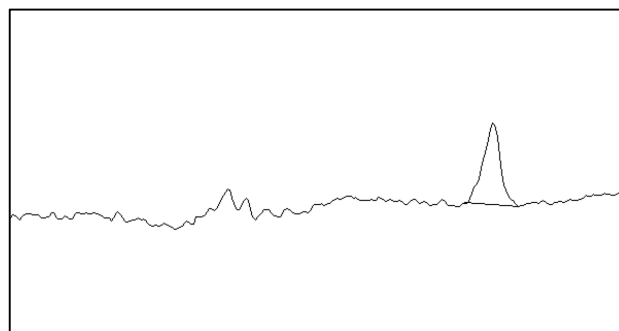


Figure 6. Peak of a single sample. Using the line tool to close off the bottom of the peak before wand tool selection.

Results

As per section 2.5, the following figures detail the Western Blot test protocol scans.

Figures include RhoA, negative control and alpha tubulin scans. Figures 11 and 12 denote the final scans used for densitometry and statistical analysis.

3.1 Western Blot Scans

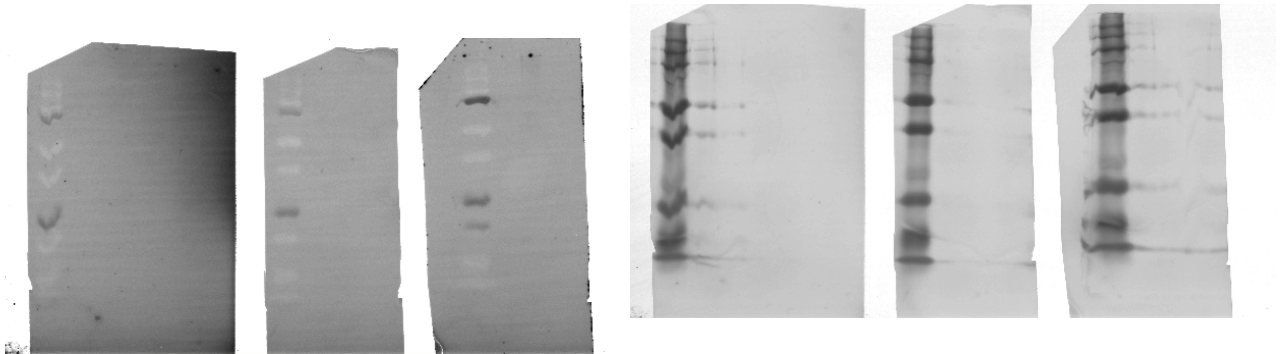


Figure 7. Test run Western Blot 1. Membranes left to right; negative control, RhoA with milk powder, RhoA with BSA, negative control, alpha-tubulin with milk powder, alpha-tubulin with BSA. Inadequate amount of protein loaded, RIPA+PI used in sample solution, assumed to be reason for blank protein lanes. Lanes left to right; ladder, gap, sample protein. This is the same across all membranes.

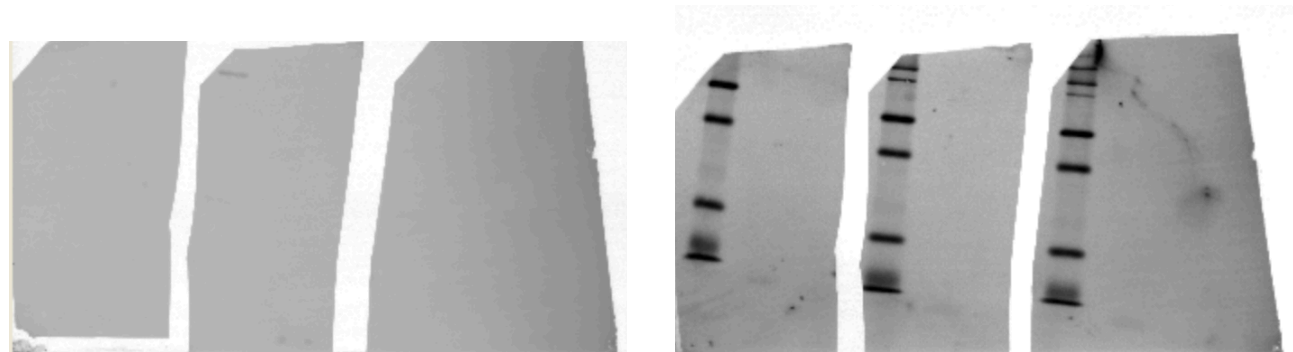


Figure 8. Test run Western Blot 2. Membranes left to right; negative control, RhoA with milk powder, RhoA with BSA, negative control, alpha-tubulin with milk powder, alpha-tubulin with BSA. Inadequate amount of protein loaded assumed to be reason for blank protein lanes. Membranes were also wonky, as shown by the ladders on the ladders on the three right hand side membranes.

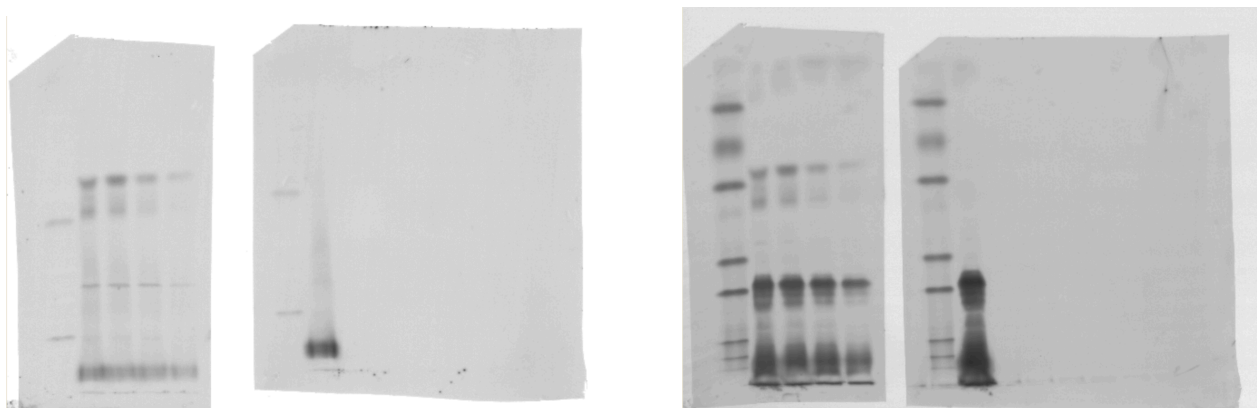


Figure 9. Test run Western Blot 3. Membranes left to right; RhoA, negative control, alpha tubulin, negative control. Lanes left to right for both membranes; ladder, 40µg, 30µg, 20µg, 10µg, ladder, 30µg. Smudging occurred, thought to be due to issues with electrophoresis. Negative control is showing protein lane when it should be blank.

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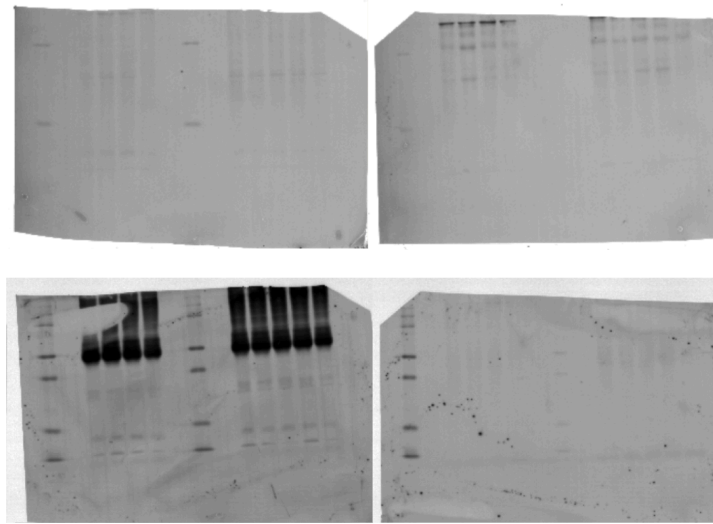


Figure 9. Test run Western Blot 4. Membranes left to right; RhoA, negative control (above), alpha tubulin, negative control (below). Lanes left to right; +/+, +/+, +/+, +/-, +/-, +/-, -/-, -/-, -/-.

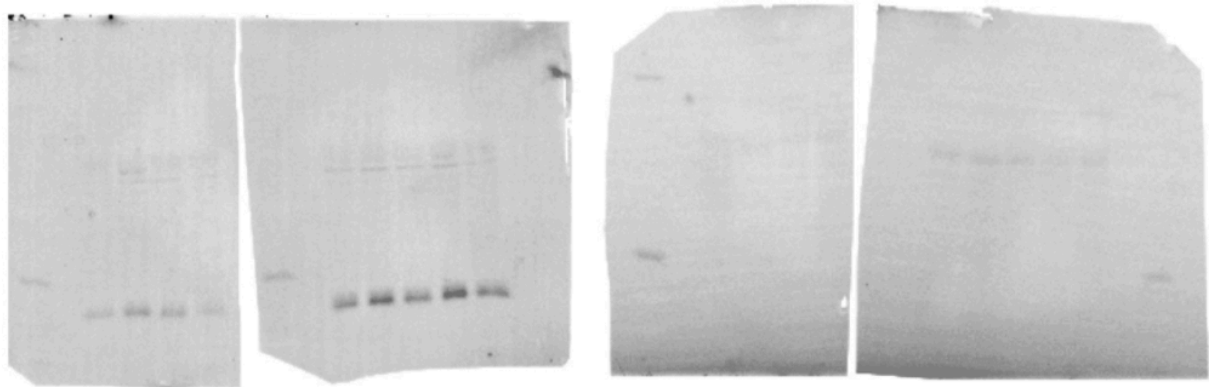


Figure 11. RhoA Western Blot membrane scan. Scanned at 473nm, FITC channel. Membrane one is scanned for RhoA, membrane three and four is negative control. Membrane one lanes indicate, ladder, gap, +/+, +/+, +/+, +/-, +/-, +/-, -/-, -/-, -/-. This is repeated in membranes three and four. RhoA is 22kDa, and so is indicated by closeness to the 25kDa ladder point.

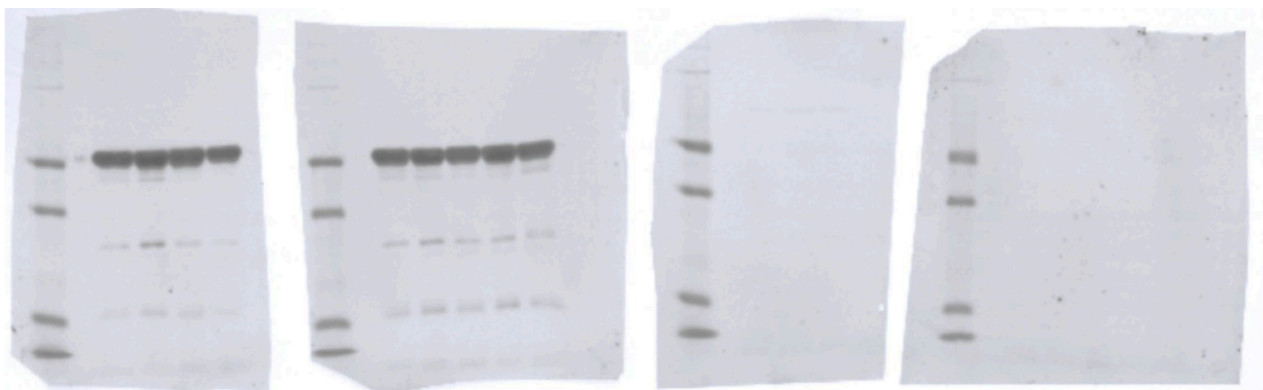


Figure 12. Alpha-tubulin Western Blot membrane scan. Scanned at 635nm, CY5 channel. Membrane one is scanned for alpha-tubulin, membrane three and four are negative controls. Membrane one lanes indicate, ladder, gap, +/+, +/+, +/+, +/-, +/-, +/-, -/-, -/-, -/-. This is repeated in membranes three and four. Alpha-tubulin is approximately 50kDa, and is indicated on the scan next to the 50kDa ladder band.

3.2 Descriptive Statistics

Densitometry analyses were conducted in order to obtain relative densities of RhoA across genotypes. Higher relative densities indicated increased levels of RhoA protein in the tissue sample. Wild types showed the lowest relative density of RhoA ($M = .49$, $SD = .15$), followed by heterozygous SERT KO animals ($M = 1.02$, $SD = .63$), followed by homozygous SERT KO animals showing the highest level of RhoA protein ($M = 1.54$, $SD = .18$).

Table 1. Relative densities of RhoA protein in wild type, heterozygous SERT KO, and homozygous SERT KO rats.

Wild Type ++	Heterozygous +/-	Homozygous -/-
0.38	0.32	1.33
0.66	1.21	1.68
0.44	1.53	1.59

3.3 Inferential statistics

A one-way ANOVA was conducted to assess differences in alpha-tubulin (loading control) percentages. No significant differences were established across wild types, heterozygous or homozygous rats ($F(2, 3.88) = 4.71$, $p = .092$), indicating loading across all lanes was equal, and variances in RhoA were not attributed to changes in protein loading. A one-way ANOVA was conducted to assess relative density differences of RhoA across genotypes. A significant difference in RhoA levels was found ($F(2, 3.59) = 25.1$, $p = .008$). A Levene's test was conducted to measure equality of variances and was non-significant ($F(2, 6) = 4.73$, $p = .058$), therefore, equal variances were assumed. Furthermore, a post-hoc Tukey test was conducted to analyse which genotypes showed significant differences in relative densities. Tukey test indicated significant differences in RhoA densities between wild type

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and homozygous genotypes ($p < .01$), however non-significant differences were found between wildtypes and heterozygous, and heterozygous and homozygous genotypes ($p > .05$).

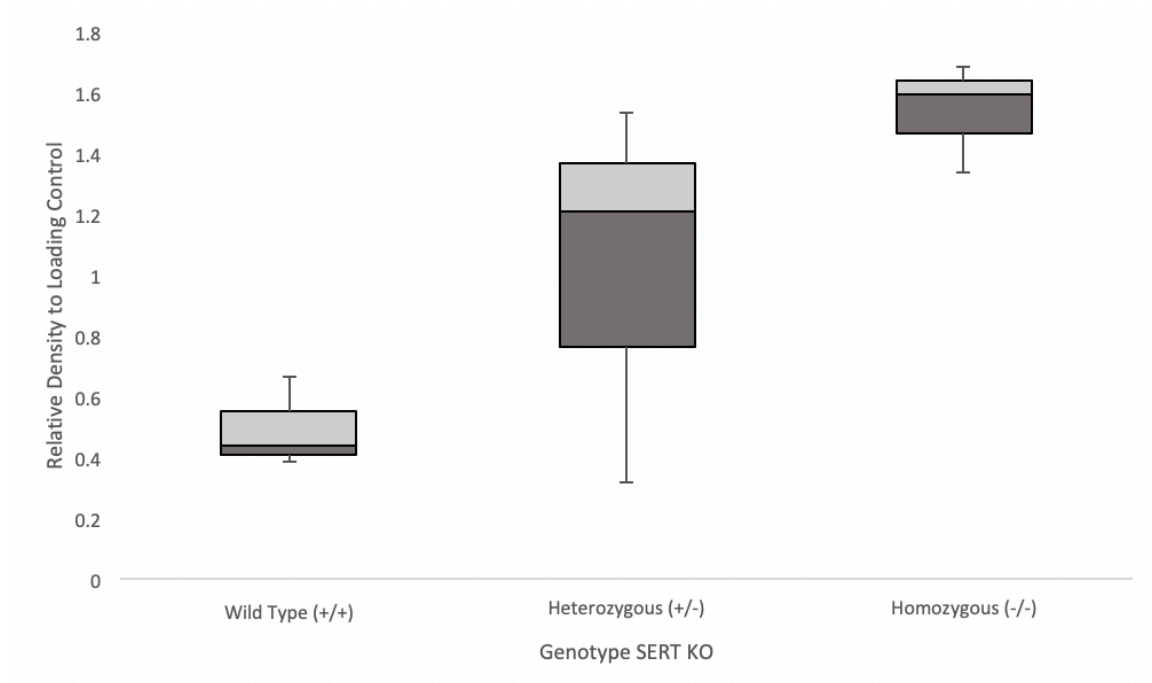


Figure 13. Box and Whisker plot for relative densities of RhoA relative to alpha tubulin. From left to right; Wild Types, Heterozygous SERT KO, Homozygous SERT KO. Significant effects between wild types and homozygous SERT KOs.

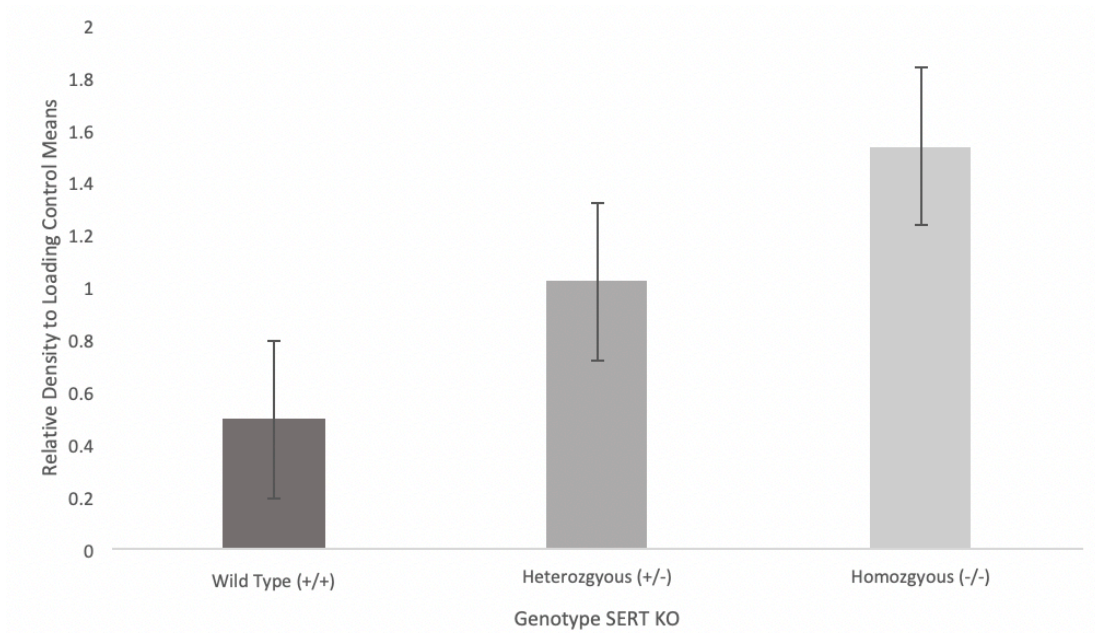


Figure 14. Bar chart for mean relative density of RhoA against alpha-tubulin loading control. Left to right; Wild types, Heterozygous SERT KO, Homozygous SERT KO. Significant effects between wild type and homozygous SERT KOs.

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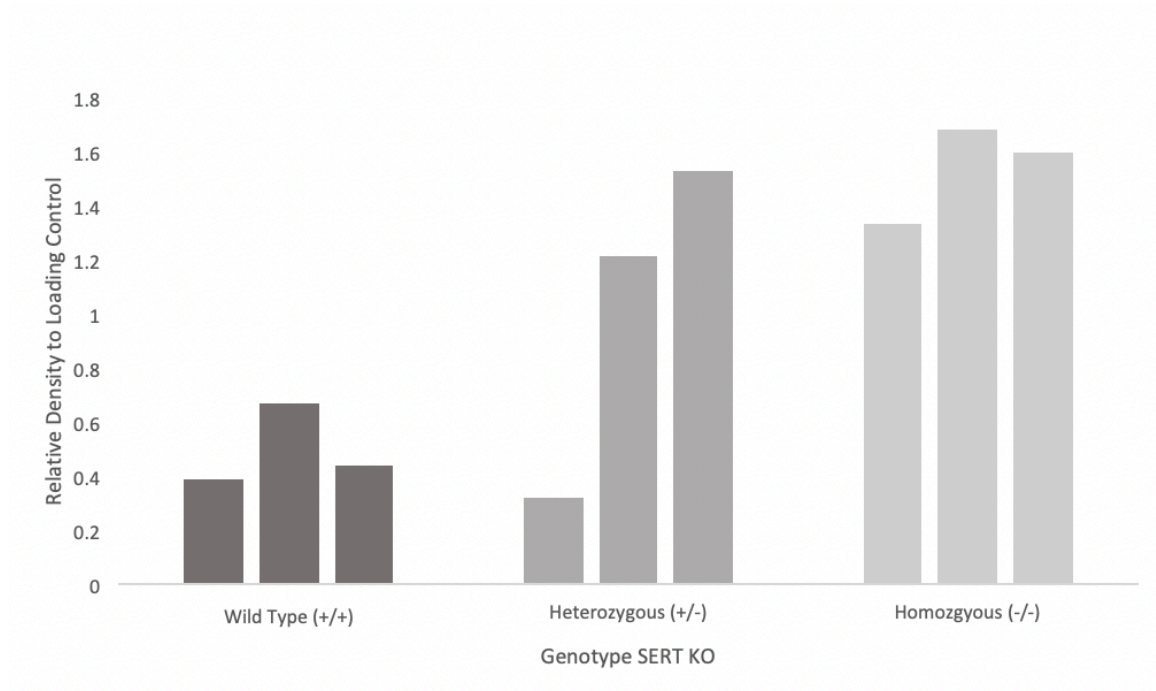


Figure 15. Bar chart for relative densities of RhoA protein of each tissue sample. Higher bars indicate higher relative densities of RhoA. Left to right; wild types; sample 1, sample 2, sample 3, heterozygous SERT KO rats; sample 4, sample 5, sample 6, homozygous SERT knockout rats; sample 7, sample 8, sample 9.

Discussion

The effect of 5-HT on neuropsychiatric disorders has long been investigated. The SERT KO rat has been used as an animal model for anxiety and depression to investigate the effects of 5-HT. The protein RhoA is activated by processes related to 5-HT mediated receptors; 5-HT₄/5-HT₇. RhoA is implicated in multiple processes within the brain and is thought to play a role in synaptic plasticity and the development of neuropsychiatric disorders. The current study aimed to develop an optimised Western Blotting procedure in order to assess the levels of RhoA in the frontal cortex. Furthermore, the study aimed to investigate differences in levels of RhoA protein using a SERT KO rat model.

Results from this study demonstrated rats with homozygous SERT KO genotypes had elevated levels of RhoA protein compared to wild type rats (see figure 13), which supports our hypothesis. Research illustrates that SERT KO animals have increased levels of extracellular 5-HT (Homberg et al., 2007). Due to our SERT KOs showing increased levels of RhoA protein, this may indicate that 5-HT has regulating effects on RhoA. However, our

hypothesis was not supported with heterozygous SERT rats showing no significant differences in RhoA protein levels among genotypes. The lack of significant differences observed in RhoA protein levels between the SERT genotypes cannot be attributed to inconsistencies in protein loading, given no differences of alpha tubulin loading control were present (~20µg each). Thus, we can confirm differences in RhoA protein levels is attributed to the genotype of the animal. However, lack of significant differences could be accredited to the small sample size. Nevertheless, considering our small sample size, finding a significant difference between wild types and homozygous SERT KO rats indicates the possible importance of 5-HT in the regulation of RhoA.

Sample number four (+/-) is remarkably lower than samples five and six (+/-) in the heterozygous SERT KO genotype group (see figure 15). It is possible environmental factors or experiences of rat number four may be implicated in why such low levels of RhoA protein were observed. Heterozygous SERT KO animals have a higher sensitivity to stressful events compared to wild types (Chaouloff, 1999). If an animal or human is exposed to long periods of stress, increases in corticosteroids and overall decrease in 5-HT will occur (Chaouloff, 2000). It is possible sample number four may have been reared in a more stressful environment/litter than those of the other rats. Due to sample four being at heterozygous SERT KO animal it will be more susceptible to stressful environments. 5-HT mediates the effect of RhoA. If sample four was raised in a naturally more stressful environment in which their levels of extracellular 5-HT was depleted, this may explain the decrease in RhoA compared to other samples within the genotype.

Our results indicate increases in levels of 5-HT due to the SERT KO genetic modification, upregulates the levels of RhoA protein in the frontal cortex. Increases in RhoA may lead to neurochemical changes in the brain implicated in developmental and behavioural changes in an individual. RhoA mediates several neuronal processes such as dendritic spine

retraction and cytoskeletal rearrangement (Manser, 2005). If these processes are disrupted or overexerted this may lead to neurological, phenotypical and behavioural changes in an individual.

RhoA contributes to cytoskeleton regulation and cell adhesions and may influence the size and morphology of dendrites and cell migration. Elevated RhoA protein levels observed in the SERT KO may be associated with widespread alterations in the brain, given the integral roles of cell migration and dendritic formation/morphology in neurodevelopment. The rearrangement of cytoskeletons has regulatory functions in various cellular processes such as morphology and migration (Ridley, Frampton, 2005). If levels of RhoA are disrupted this may have detrimental effects of the development of the brain through abnormal dendrite morphology and cell migration. During neurogenesis stages of development, cells migrate to their respective areas of the brain. However, if cell migration is interrupted, then axons and dendrites may not migrate to the correct areas. During prenatal periods, if a mother experiences stress related events, migration of cells can be delayed (Fine, Zhang, Stevens, 2014). Prolonged periods or chronic stress can lead to depletion of 5-HT (Chaouloff, Berton, Mormede, 1999). It is plausible decreases in 5-HT may lead to decreases in RhoA protein levels, and therefore depleted RhoA activation. In the absence of sufficient RhoA protein levels cell migration not occur correctly. Inadequate amounts of RhoA during development may initiate dysfunctions in cell migration processes, leading to the improper placement of cells throughout cortical structures.

Extensive research has demonstrated the importance of 5-HT in the development of neuropsychiatric disorders through use of SERT KO models and SERT blocker medications (Terry, Buccafusco, Wilson, 2008). Nevertheless, serotonergic transmission may be only one of many mechanisms playing a role in synaptic plasticity. Processes mediated by RhoA protein may be implicated in changes of synaptic plasticity and may play a role in the

development of neuropsychiatric disorders. Synaptic plasticity is the ability of neurons to bring about changes within connections between neural networks in the response to chemical, structural, functional and mechanistic abnormalities in the brain (Honer et al, 1999). It is suggested synaptic plasticity may have effects on the development of neuropsychiatric disorders, i.e., decreases in the ability for the brain to adapt its neural networks and connections may be a risk factor for the development of these disorders.

RhoA's regulating effects on dendritic spine growth may be implicated in synaptic plasticity and neuropsychiatric disorder development. As stated in the literature, 5-HT₄ and 5-HT₇ receptors are implicated in the activation of RhoA. Increased 5-HT will result in increased 5-HT₇ and 5-HT₇ receptor activation, resulting in the surge of RhoA protein levels (Tolias, Dunman, Um, 2011). Following, the activated RhoA proteins will engage in their respective processes. If there is over activation of RhoA proteins this may lead to the over exaggeration of these processes such as dendritic spine retraction, cytoskeletal rearrangement and cytokinesis. Whilst spine retraction is a necessary process, if this is overcompensated, it may have maladaptive effects in the brain (Honer et al., 1999). Dendritic spine retraction may lead to a disconnection between neural networks within the brain. Neural networks that are disrupted or dysregulated have been implicated in the development of schizophrenia as individuals who suffer from the disorder demonstrate a fundamental degradation in basic neural network activity (Light et al., 2006). The dysregulation of the above processes may lead not only to dysfunctions in synaptic plasticity but may cause a functional disconnectivity within the brain, which has been implicated in the development of depression (Liao et al., 2013). This demonstrates the possible function of the RhoA protein on synaptic plasticity as well as the development of neuropsychiatric disorders.

Cytoskeletal rearrangement helps to regulate various cellular processes such as morphology, migration and cytokinesis (Manser, 2005). RhoA participates in the regulation

of actin cytoskeletons and a crucial balance of RhoA is necessary in order to ensure actin is regulated appropriately (Manser, 2005). Cytoskeletal actin contributes to cell membrane repair and if it is disrupted can cause dysfunctions in repair processes (Kim, Kwon, 2015). The rearrangement of cell skeletons is crucial to cellular movement and motility. RhoA is mediating the effects of cytoskeletal rearrangement and actin filaments, therefore being implicated in the changes in cell morphology. If excessive RhoA is present, this could result in detrimental effects such as inability to repair cells or induce maladaptive morphologies, or decreased motility. This may lead to further functional disruptions and disconnectivity throughout the brain, further emphasising the importance of RhoA in neural development.

Increases in RhoA protein levels may also lead to enhanced sensitivity to drugs of abuse. The retraction of dendritic spines and decrease in cytoskeletal rearrangement may lead to a depletion in synaptic plasticity. This may lead to the faster acquisition of addictive behaviours, as the brain may not be able to compensate for the pharmacological changes occurring in the brain during these events, due to the RhoA induced decrease in synaptic plasticity. MDMA treated animals in mRNA sequencing data demonstrated downregulation of RhoA compared to clean animals (see figure 1). Long term effects of MDMA administration lead to depletion of extracellular 5-HT and structural damage and downregulation of 5-HT receptors (Lizarrage et al, 2014). If an individual's neural networks are more rigid, the structural damage caused by chronic usage of drugs of abuse may not be repairable, or the brain may not be able to adapt to mediate these effects. Implying the importance of RhoA.

4.1 Limitations and Future Directions

Western Blotting protocol needed testing multiple times. Due to optimisation taking longer than expected more experimental procedures were not able to occur. Additionally, it is unclear if the absence in RhoA differences between heterozygous SERT KO and the wild

type and homozygous SERT KO rats is indeed a reflective result of differences in SERT functionality, or simply an outcome of the limited sample size.

In relation to the SERT KO model, the heterozygous animals can be compared to the SERT short allele implicated humans, as both polymorphisms demonstrate decreased SERT expression. However, SERT expression is only partially decreased, unlike the homozygous SERT KO rat. The short SERT allele has been observed as increasing the risk for multiple mental health related conditions such as depression, anxiety, and stress-related substance abuse (Glenn, 2011). Functionally, the short allele is associated with reduced transcriptional expression of SERT compared to the long allele. However, in our results no differences were seen in heterozygous SERT KO animals, therefore it remains tentative whether our results are generalisable to the human population. Due to a lack of observed differences in heterozygous SERT KO animals, we cannot definitively say that in humans a decrease in SERT expression may lead to RhoA increases and the functional changes mentioned above. Therefore, emphasising the importance of further research into the effect of SERT polymorphism RhoA, as this study was regarded as a pilot.

Further optimisation of the RhoA western blotting protocol may be necessary, including the addition of more samples. In future studies, it would be advantageous to assess RhoA levels earlier in life. RhoA could also be measured at differing age points in order to assess if RhoA changes across developmental periods and assess the possible changes and mechanisms in neurodevelopment. Animals could be taken at post-natal-day 7, 21, and 60. These particular time periods in rats can be related to neonatal, periadolescent, and young adulthood in humans and therefore can be generalised back to the human population (Sengupta, 2013). Through testing RhoA levels at different age points we may be able to see and understand the developmental effects RhoA has on the brain and this may coincide with the investigation of dendritic spines. As discussed above, RhoA is implicated in spine growth

and retraction. A future study could include research into spine morphology in animals presenting with differing levels of RhoA, such as the SERT KO model, and to examine if this has an effect of the brain development of these animals.

4.2 Conclusions

In conclusion 5-HT is a clear mediator of RhoA protein density within the frontal cortex. Increases in 5-HT levels lead to upregulation of RhoA protein levels. RhoA may be a crucial protein implicated in the development of the brain, synaptic plasticity and development of neuropsychiatric disorders. RhoA and its regulating effects on dendritic spine retraction, cytoskeletal rearrangement and cell morphology may lead to behavioural alterations such as increases in depressive and anxiety like symptoms, and enhanced sensitivity to drugs of abuse in genetically compromised SERT KO animals. It is plausible increased RhoA levels may be identified as a risk factor for synaptic plasticity reduction and neuropsychiatric disorder development. However, further optimisation of the RhoA western blotting protocol is necessary, including expansion of sample size, testing animals in earlier stages of life, as well as variations in developmental time periods.

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Appendix A

Tables for RhoA and Alpha tubulin areas, percentages and relative densities

Table 1. RhoA areas, percentages and relative density to alpha-tubulin

Sample/Type	Area	Percentage	Relative Density
Sample 1 (WT +/+)	355.63	4.39	0.38483121
Sample 2 (WT +/+)	712.82	8.80	0.664702327
Sample 3 (WT +/+)	471.58	5.82	0.43994255
Sample 4 (Het +/-)	240.26	2.96	0.321688185
Sample 5 (Het +/-)	975.36	12.04	1.210822772
Sample 6 (Het +/-)	1355.94	16.74	1.530078625
Sample 7(Hom -/-)	1187.36	14.70	1.334456383
Sample 8 (Hom -/-)	1614.89	19.93	1.6813159
Sample 9 (Hom -/-)	1188.18	14.67	1.594541698

Table 1. RhoA areas, percentages and relative density to alpha-tubulin

Sample/Type	Area	Percentage
Sample 1 (WT +/+)	5546.79	11.41
Sample 2 (WT +/+)	6437.15	13.24
Sample 3 (WT +/+)	6437.15	13.23
Sample 4 (Het +/-)	4482.43	9.22
Sample 5 (Het +/-)	4834.96	9.94
Sample 6 (Het +/-)	5319.55	10.94
Sample 7(Hom -/-)	5340.84	10.98
Sample 8 (Hom -/-)	5765.38	11.86
Sample 9 (Hom -/-)	4472.79	9.20

Appendix B

Statistical Tables for Relative RhoA Protein Density and Alpha-Tubulin Loading Control.

Table 1. One-Way ANOVA of RhoA Relative Protein Density

	F	df1	df2	p
Relative Densities	25.1	2	3.59	0.008

Table 2. Group Descriptives of RhoA Protein Density

	Type	N	Mean	SD	SE
Relative Densities	WT (+/+)	3	0.496	0.148	0.0856
	Het (+/-)	3	1.021	0.626	0.3615
	Hom (-/-)	3	1.537	0.181	0.1042

Table 3. Test for Equality of Variances (Levene's)

	F	df1	df2	p
Relative Densities	4.73	2	6	0.058

Table 4. Tukey Post-Hoc Test – Relative Densities of RhoA Protein

		WT (+/+)	Het (+/-)	Hom (-/-)
WT (+/+)	Mean difference	-	-0.524	-1.040*
	p-value	-	0.292	0.038
Het (+/-)	Mean difference		-	-0.516
	p-value		-	0.302
Hom (-/-)	Mean difference			-
	Hom (-/-)			-

Note * $p < .05$, ** $p < .01$, *** $p < .001$

Table 5. One-Way ANOVA of Alpha-Tubulin Loading

	F	df1	df2	p
Relative Densities	4.71	2	3.88	0.092

Table 6. Group Descriptives of Alpha Tubulin Loading.

	Type	N	Mean	SD	SE
Relative Densities	WT (+/+)	3	12.6	1.055	0.609
	Het (+/-)	3	10.0	0.864	0.499
	Hom (-/-)	3	10.7	1.355	0.782

Table 7. Test for Equality of Variances (Levene's)

	F	df1	df2	p
Relative Densities	0.476	2	6	0.643

Appendix C

RhoA Western Blotting Protocol.

DAY ONE

Gel Preparation

Add parafilm to the bottom of your plates before placing in gel-setting apparatus. This is to prevent the gel from leaking out while setting.

Prepare a 10% Separating Gel (enough for 2 gels)

- 8ml ddH₂O
- 5ml 1.5M Tris 8.8
- 200ul 10% SDS
- 6.66ml Acrylamide
- 100ul 10% APS
- 10ul TEMED

Pipette in between plates using serological pipette and wait at least 30 min to set. To prevent gel from drying out, cover top with water saturated Butanol while gel is setting (using isopropanol will actually dry it out further)

Tip off Butanol before adding Stacking Gel

Make 4% Stacking Gel in meantime but add the catalysing components (APS, TEMED) only right before adding the stacking gel.

Note: You can make the separating gel in advance – once gel has set, wrap up in drenched paper towel (using running buffer – 1%) Put it into a bag or glad wrap and into the fridge so that it doesn't dry out. The stacking gel should always be made fresh.

4% Stacking Gel

- 6.1ml ddH₂O
- 2.5ml 0.5M Tris 6.8
- 100ul 10% SDS
- 1.33mL Acrylamide
- 50ul 10% APS
- 10ul TEMED

Once separating gel is set, fill plate with stacking gel to the top. Insert the 1.5mm comb directly into the gel to generate the wells/lanes, but be careful not to force it down to avoid acrylamide splashing on you.

Use paper towel for this

Sample Prep

From the results of the BCA Assay, use enough of the sample to have 20ug of protein per lane – make 30ul to be safe

Add equal amount of 2x Laemmli Buffer (that now has 50ul B-mercaptoethanol in it), to make a 1:1 ratio of sample: Laemmli

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Denature by heating samples at 95° for 5 minutes

Centrifuge the samples to bring all the protein to the bottom of the tube, then place back onto ice to prevent protein degradation

*all samples, APS, RIPA etc. should be kept cold/on ice at all times

SDS-PAGE

Once stacking gel is ready, prepare for electrophoresis.

Put Gels into clamps facing inwards, with a buffer dam or another gel placed on the other side to form a compartment

Fill between gels with 1x running buffer, and fill tank to either 2 gel or 4 gel mark depending on number

- Use 1x SDS-PAGE running buffer for this (from 10x solution: 100ml 10x SDS PAGE, 900ml of ddH₂O)

Remove comb carefully to not destroy the lanes and let the lanes/wells fill up with running buffer

Add precision Plus protein dual colour standard (the ladder) into the first well

Made up of: 2x Laemmli, RIPA + PI, 1ul colour standard

- eg. For 10ul total: 5ul 2x Laemmli Buffer, 4ul RIPA + PI, 1ul colour standard

Leave gap for second well

Add 20-30ul from each sample into the other lanes, to load 20ug of protein. Normalise volume of sample if necessary

Make sure samples are pipetted carefully into the wells and do not spill over to adjacent walls

Put the tank lid on and make sure electrodes are fitted onto the pronged clamps. Begin electrophoresis for 1.2-2 hours at 120V

Transfer

Make Transfer Buffer

- 6.06g Tris Base
- 28.8g Glycine

Mix Salts with 1.6L of dH₂O to dissolve

Add remaining 400ml of methanol and stir to dissolve

MAKE SURE THE TRANSFER BUFFER IS **COLD**, you should make this while electrophoresis is taking place

- Put into the -70° for 20-30min (be careful to not let it freeze)

Prepare your equipment

Fill bucket with ice and put transfer container into it. Fill container with transfer buffer up to 'blotting' line

Put ice pack into the tank

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Equilibrate all transfer items – cassettes, fibre pads, filter paper onto COLD transfer buffer

Soak PVDF membrane in methanol 10 minutes before electrophoresis is complete and also equilibrate that in transfer buffer

DON'T YOU DARE TOUCH THE MEMBRANE – USE TWEZERS

Add more buffer to ensure satisfactory equilibration

Once Electrophoresis is complete, remove the short plate and stacking gel **CAREFULLY** with the green spoon, and make your transfer sandwich.

Order Should Be

BOTTOM	Grey Side of Cassette Fibre Pad 3 MM Filter Paper Electrophoresed Gel PVDF Membrane 3 MM Filter Paper Fibre Pad
TOP	Clear Side of Cassette

Ensure there are no air bubbles by using the roller. In this orientation, the membrane is **FACE DOWN**, which will be important later on

Once sandwich is made, place your cassette into the holder in the transfer buffer. Make sure grey side of cassette is to the back, and white side is to the front.

If running two gels, make two sandwiches. If running one gel, make a black cassette with everything besides a gel and membrane

Move bucket containing the transfer tank into the cold room (4C) and run transfer for 17 hours at 20V OR for 3-4 hours at 70V

DAY TWO

Remove the membrane from the sandwich, and snip the top left corner of the membrane to indicate that this is the **FACE DOWN** side

Place it **FACE UP** into a click box and add 1x TBS. Wash for 5 minutes (you will know the membrane is face up because corner is on the top right side and ladder is to the right)

Block membrane 5% Milk Powder in 1x TBST for 1 hour at RT with gentle rocking (15-25RPM)

*Filter BSA using a 0.22um filter to prevent large particulates which will attach onto the membrane and make your results look like shit when you scan it

After blocking, incubate the membrane with the primary antibody (RhoA1B12) overnight at 4°C.

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RhoA Antibody concentration: 1:2000, In 1% Milk Powder in 1xTBST (V = 5ml)
Incubate with gentle rocking (15-25RPM) – Cold room

Note: You should also have a negative control membrane here too. This can be done by either cutting the membrane in half and using one for the primary antibody and another without the primary OR having two separate membranes

For negative control – incubate overnight at 4° with 1% Milk Powder in 1xTBST without the primary

Coomassie Staining

A Coomassie stain can also be performed on this day with the gel to determine whether all the proteins transferred onto the membrane (or not)

Place gel in box and add SDS PAGE Gel Fixative (40% dH₂O, 50% methanol, 10% acetic acid). Let rock (15-25RPM) for 15 minutes, then add Coomassie Brilliant Blue Powder
Incubate in the Coomassie solution for 4-hrs overnight

Remove fixative solution and add in the washing solution and add in the washing solution (same chemicals but a higher concentration of water and lower concentration of methanol and acetic acid): 67.5% dH₂O, 7.5% acetic acid, 25% methanol

Wash in this mixture two or three times, with fresh addition every time. It can be left to rock (n roll) overnight

This will wash off the blue stain and will show if some heavier proteins remained in the gel and did not transfer

DAY THREE

Remove primary antibody mixture from click box, and wash membrane 3x 5 min in 1xTBST

Then incubate both primary membrane and negative control membrane with secondary antibody

- Use Alexa 488 (Gt x MS) at 1:4000 dilution in 1% Milk Powder in 1xTBST (V = 5ml)

Incubate for two hours at room temperature but COVER the box by putting it into a black bag. Rock (n roll) at 15-25 RPM at room temperature

Once complete, wash 3x 5 min in TBST before scanning

- Keep covered with black bags to avoid light

Scanning

Using the FLA-5100 Fluorescent Image Analyser, scan the membrane

- Place membrane FACE DOWN – If using 488 (as done here) scan using 473 lasers and FITC channel

Also scan for dual colour Ladder protein standard – scan at 635nm (Cy5 channel) to obtain steps in the ladder

Loading Control

Incubate once more to reprobe for loading control detecting (Alpha-tubulin)

A-tubulin 1:10,000 concentration – incubate for 1 hour in 1% Milk powder in 1xTBST

Then incubate with secondary AB, Donkey anti-mouse Cy5, for 2 hours at concentration of 1:4000. Scan membrane at 635nm (Cy5 laser).

A-tubulin loading across all samples should be the same

Stripping the membrane

If stripping and re-probing is required, prepare a harsh stripping buffer:

Harsh Stripping Buffer

- 20ml SDS 10%
- 12.5ml 0.5M Tris HCl pH 6.8
- 67.5ml dH₂O
- 0.8ml B-mercaptoethanol added under the fumehood

Prepare buffer and mix well

Add membrane to a plastic box and cover membrane with buffer

Incubate up to 45 minutes with mild agitation, under the fume hood

Remove buffer (put in one of those disposable liquid bottles), and rinse membrane in the box with dH₂O for 1-2 minutes

Then wash in TBST for at least 5 minutes. Wash as long as possible because if there are still traces of B-mercaptoethanol it will damage antibodies

Block with 5% Milk powder and continue with protocol as above